# Technology Stack

**Date:** 14 July 2026 | **Team ID:** | **Project Name:** Emotion Detection and Learning Support Engine

The Emotion Detection and Learning Support Engine is developed using a modern AI and web application technology stack. Each technology is selected to support efficient emotion detection, personalized learning guidance, data management, visualization, and future scalability.

| S.N. | Architecture Component / Layer | Technology Chosen | Justification / Purpose |
|---|---|---|---|
| 1 | Frontend / Client-side | Streamlit (Python) | Provides an interactive, lightweight, and user-friendly web interface for users to enter text and view predictions, guidance, and analytics. |
| 2 | Backend / Server-side | Python 3.11, TensorFlow, Keras | Handles text preprocessing, emotion prediction using the BiLSTM model, business logic, and learning guidance generation. |
| 3 | Database / Data Storage | CSV Files (Emotion Logs), Pickle (.pkl) Files | Stores emotion prediction history, tokenizer, and label encoder for model inference. |
| 4 | Cloud Hosting / Deployment | Streamlit Cloud (Deployment Ready) | Enables easy deployment and online accessibility of the application. |
| 5 | Version Control & CI/CD | Git and GitHub | Used for source code management, version control, and project collaboration. |
| 6 | Third-party APIs / Other Tools | Google Gemini API, NumPy, Pandas, Matplotlib, Scikit-learn | Gemini API generates AI-powered learning guidance; NumPy and Pandas support data processing; Matplotlib provides analytics visualization; Scikit-learn supports preprocessing utilities. |